[Teleprzyjaciele_IBM Z Datathon 2026 Project]

Selected Theme: Real-Time AI for Critical Decisions (Cross-cutting: AI Secured, AI for Good)

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1. Problem Statement & Community Impact

Online payment fraud disproportionately harms seniors and non-technical users under intense psychological pressure. According to FBI IC3 reports, elder fraud losses exceed $3.4 billion annually [1], with social engineering, lookalike domains, and urgent bank impersonation driving the surge. Victims are coerced into authorizing irreversible transfers within minutes. Current banking defenses fail because they evaluate signals in silos: a DNS check, a session token rule, or an isolated transaction limit. Attackers exploit these gaps: a domain registered two hours ago, a subtle TLS/ASN session shift, and an urgent transfer to a new payee appear benign in isolation, yet together represent an active account takeover. Protecting life savings requires real-time, cross-layer signal correlation before funds leave the account.

2. Proposed Solution & Innovation

We propose a prototype of an inline, millisecond-scale fraud scoring engine designed for IBM Z, using a Pipes-and-Filters architecture with an intelligent fast-path:

- **Fast-Path:** Clearly benign events (whitelisted domains with regular transfers) and known blacklisted domains (data from PhishTank) bypass deep inference, preserving high transaction throughput.
- **Algorithmic Domain Analyzer:** Evaluates suspicious URLs in real time by detecting homograph/Punycode attacks (via homoglyph datasets based on Unicode UTS #39) [8], typosquatting (Levenshtein distance against the Tranco Top Sites ranking) [6], and algorithmic domain generation (Shannon entropy and character n-grams).
- **Transaction Analyzer:** Employs recurrent neural network models consisting of LSTM/GRU layers to analyze sequential transaction histories and detect behavioral anomalies. Following the IBM Z ONNX deployment pattern [7], our custom-trained models are exported to the ONNX format for low-latency execution.
- **Correlation Engine & Reason-Code Explainability:** Computes a hybrid composite score (weighted sum, critical override triggers, and a synergy penalty for co-occurring medium risks) to issue a definitive action: *Allow*, *Step-Up Verification*, *Delay*, or *Block*. Instead of opaque blocks, the engine aggregates granular reason codes from active shields (e.g., combining domain spoofing alerts with transactional anomaly metrics) to dynamically assemble clear, contextual warnings for seniors alongside UI friction that breaks psychological coercion.
- **Future Roadmap (Session Analyzer):** The modular pipeline is architected to ingest in-session telemetry (JA3/JA4 TLS fingerprint shifts and IP/ASN geolocation jumps) to neutralize Adversary-in-the-Middle (AiTM) session hijacking in subsequent iterations.

3. Data & AI/ML Methodology

We combine verified domain repositories (Tranco, PhishTank, and homoglyph reference datasets) with transactional data from the PaySim dataset [5]. Combined multi-stage elder fraud scenarios—pairing lookalike/phishing domain access with sequential account-drain transfers—are explicitly synthesized. To prevent circular evaluation, models are validated using time-based splits and

stress-tested against unseen attack topologies. Our methodology includes a formal ablation study contrasting Domain-only heuristics, Transaction-only (LSTM/GRU), and Hybrid Correlation performance across PR-AUC, Recall, False Positive Rate (FPR), and end-to-end inference latency under target real-time banking SLAs [4].

4. Why IBM Z & LinuxONE?

Fraud scoring must occur directly inside the transactional pipeline; offloading inference to remote x86 clouds forces institutions to skip AI checks during peak loads to meet SLAs. We build and benchmark our prototype on the IBM LinuxONE Community Cloud as a direct stepping stone toward production deployment on **IBM z16/z17**. Deploying ONNX Runtime natively on LinuxONE co-locates LSTM/GRU inference and algorithmic domain shields with core transactional data, bringing scoring latency down to the single-digit millisecond range. In a target production environment, the sequence model directly exploits integrated on-chip AI acceleration on **IBM z16/z17 (Telum/Telum II)** [4], while hardware encryption and **IBM Secure Execution (TEE)** [2] isolate sensitive senior citizen PII using quantum-safe cryptography [3].

References

[1] Federal Bureau of Investigation. Elder fraud report 2023. Technical report, Internet Crime Complaint Center (IC3), 2023.

[2] IBM Corporation. Ibm secure execution for linux. Technical report, IBM, 2023.

[3] IBM Corporation. Quantum-safe security for ibm z. Technical report, IBM, 2024.

[4] IBM Redbooks. *Journey to AI on Z*. IBM Corporation, 2023. REDP-5723-00.

[5] Edgar Alonso Lopez-Rojas, Ahmad Elmir, and Stefan Axelsson. Paysim: A financial mobile money simulator for fraud detection. In *28th European Modeling and Simulation Symposium (EMSS)*, 2016.

[6] Victor Le Pochat, Tom Van Goethem, Samaneh Tajalizadehkhoob, Maciej Korczyński, and Wouter Joosen. Tranco: A research-oriented top sites ranking hardened against manipulation. In *Proceedings of the 26th Annual Network and Distributed System Security Symposium (NDSS)*, 2019.

[7] Suona Slafalce and Steve Martinelli. ai-on-z-fraud-detection, 2021. Repository for credit card fraud detection notebooks and ONNX models on IBM Z.

[8] Unicode Consortium. Unicode technical standard #39: Unicode security mechanisms, 2022.